RITWIK KATTA

Software Engineer Intern

+91-7483088704 | ritwikkatta7@gmail.com | Upwork Profile | Linkedin | Github

EXPERIENCE

Software Engineer (Upwork) May 2025 – May 2026

NSE Intraday Trading Platform
Python, Zerodha Kite Connect API, SQLite

- Built an **event-driven trading platform** with **8 modular components** spanning market data, strategy, execution, persistence and risk.
- Processed live market data for **50 NIFTY instruments** across NSE.
- Built a **5-year backtesting engine** modeling brokerage, taxes & slippage.
- Added **crash recovery** and **persistent risk controls** for reliability.

SEO Landing Page Automation Engagement
n8n, REST APIs, Web Scraping

- Built the frontend of an automated **SEO landing-page workflow**.
- Delivered page structure, **reusable UI components** and **REST API integration**.

ACHIEVEMENTS

- Solved **150+** DSA problems on **LeetCode**.
- Achieved a **5-Star Gold Badge** on **HackerRank**.
- Earned **Top Rated** status on Upwork for Python, API and automation work (**Upwork Reviews**).

TECHNICAL SKILLS

Languages: Python, Java, C, SQL

Backend: FastAPI, Node.js, Express, REST API design

Frontend: React

Databases: MySQL, SQLite, MongoDB, ChromaDB

Tools: Git, GitHub, Linux, n8n, Airtable

COURSEWORK SUBJECTS

- Operating System
- Computer Networks
- Object Oriented Programming
- Database Management System

EDUCATION

B.Tech. (CSE - AI) - 7.36 CGPA

Manipal Institute of Technology

2023 – 2027 | Bengaluru, Karnataka

PROJECTS

AI PR & Code Review Assistant

MERN (MongoDB, Express, React, Node.js)

- Full-stack tool that connects to **GitHub**, analyzes real PRs and flags risky changes.
- Uses **AI-generated summaries** to explain each flag.
- Built a transparent **risk-scoring formula**: diff size, sensitive files, test coverage.
- Validated on **5+** real-world repos with **40 automated tests**.
- **Server-side sorting/pagination** keep highest-risk changes surfaced first.

AI Knowledge & Document Platform

Python, FastAPI, ChromaDB

- **FastAPI** service: upload files, get answers with **page-level citations**.
- Uses **hybrid search** and AI-based re-ranking.
- Implemented a custom **HNSW search algorithm** from scratch, verified correct.
- Benchmarked against brute-force search and **ChromaDB**.
- Evaluated on **35 questions**: **0.950 MRR** with cross-encoder re-ranking.

Predictive Maintenance for Jet Engines

Python, scikit-learn, XGBoost, FastAPI

- Predicts jet engine failure from real **NASA sensor data**, comparing **3 regression models**.
- Found and fixed a **critical, safety-relevant bug** returning wrong risk levels for low-history engines.
- Tuned the threshold with **F2-weighted** evaluation, cutting missed failures by **47%**.